

SatQuery AI - Multimodal Remote Sensing Evidence Report

Autonomous Agentic Execution & Evidence Provenance Trace

Query:	"Highlight the water body referred to in the query."
Task Category:	GROUNDING
Specialist Tool:	RS Text-Guided Region Grounding Specialist
Reliability Score:	80.4% (MEDIUM TRUST)
Execution Time:	105.8 ms

1. Agent Synthesis & Visual-Language Answer

Text-Guided Region Grounding Output:

- 1. Target Entity Delineated: 'Water Body / Water Corridor'.
- 2. Bounding Box Coordinates [ymin, xmin, ymax, xmax]:
 - Normalized: [0.1328, 0.0703, 0.9668, 1.0]
 - Pixel Grid: [68, 36, 495, 512]
- 3. Grounded Region Extent: 77.53% of scene tile area.

2. Auditable Agent Pipeline Execution Trace

- **[Perception Layer]** Format & CRS validated. Mode: Single Optical Image (1 image(s)). Status: PASSED
- **[Query Engine]** Classified intent: 'GROUNDING'. Extracted entities: ['water']. Task: 'grounding'
- **[Mission Planner]** Selected tool: 'RS Text-Guided Region Grounding Specialist' with params {'target_entities': ['water']}. Generated 7 node task-graph.
- **[Specialist Tool Execution]** Executed specialist tool 'RS Text-Guided Region Grounding Specialist' with model confidence 74.8%.
- **[Evidence Fusion]** Aligned 2 evidence streams. Spatial Bounding Box aligned with query subject.
- **[Trust & Uncertainty]** Calculated Reliability Score: 80.4% (MEDIUM TRUST). Cross-model agreement: 76.1%.
- **[Discovery Engine]** Autonomous scan completed: Autonomous scan identified 1 'Beyond Your Query' discovery item(s).
- **[Answer Generator]** Final evidence-linked response synthesized in 105.8 ms.